## 4.1 Depth determination

Depth determination is a fundamental task for a hydrographer, which requires specific knowledge of the techniques and equipment involved. According to the IHO publication S-44 5th Edition, the uncertainty of depth measurements shall be estimated at the 95% confidence level for every sounding in the survey.

During the last decade, hydrographic surveying has experienced a conceptual change in depth measurement, moving from single-beam echo sounders to multibeam and LiDAR systems. The 2nd generation of swath systems delivers dense coverage of the seafloor with reduced survey time and improved accuracy.

Lead line and sounding pole were the earliest methods used for directly measuring water depth. These methods are still applied in very shallow waters, where acoustic instruments cannot be operated safely, and they remain useful as an independent check of the echo sounder calibration.